Pilot cost summary

One proposed township area, 100 monthly active participants and three operating months. All uptake and cost ranges are modelling assumptions.

TECHNICAL FEASIBILITY

Map data + GPS + a points ledger. Estimate shade and flag data gaps. Third-party vending integration is documented for compatible platforms.*4 *5

OPERATIONAL FEASIBILITY

SDP programme owner + compatible operator + funded water rewards. Confirm the actual machine, API access and full fulfilment price before a live pilot.

UPFRONT SETUP FUNDING

RM 15,279

Includes build, onboarding, launch, domain and reserve.

EXPECTED MONTHLY OPEX

RM 1,554

200 water redemptions/month, including 10% reserve.

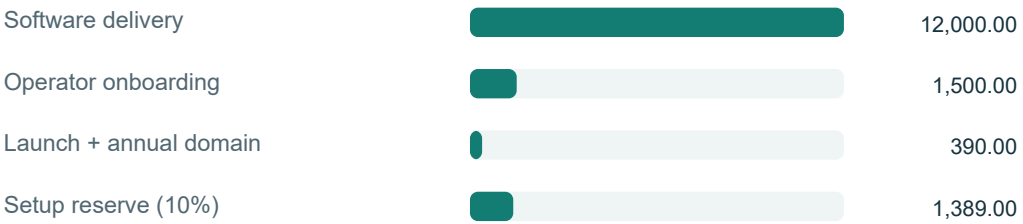
3-MONTH FUNDING CEILING

RM 20,930

Funds up to 400 water redemptions/month at base rates.

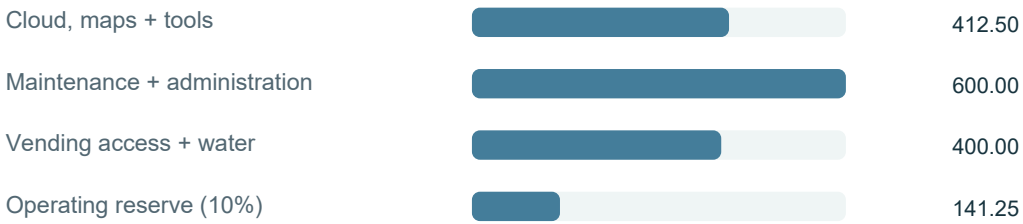
CAPEX / setup budget*

One-off base case (RM)



OPEX budget

Expected base case, RM per month



Expected value: repeat participation · traceable water rewards · evidence for pedestrian improvements

*One-off budget is not automatically accounting CAPEX. Classification and source keys are on pages 2 and 9. Prices are benchmarks; operator charges are allowances.

Assumptions | scope and evidence status

Mabe's appendix structure is reused: low/base/high, separate time periods, source notes and a combined summary. Pharmacy costs are not reused.

Input	Working assumption / qualification
Pilot setting	One township activity area and one existing compatible machine. Exact route, operator and location not secured.
Participants and duration	100 monthly active participants; 3 operating months. These are planning inputs, not forecast adoption.
Water volume	Up to 4 bottles per participant per month; 50% of maximum allocations redeemed in the expected case: 200 bottles/month.
Reward policy	Budget counts confirmed bottles, not outstanding points. Point conversion and eligibility remain product decisions. Pilot caps are proposed assumptions.
Currency and FX	All totals in MYR. USD prices converted at a planning rate of RM 4.50/USD; this is not a live exchange-rate quote.
Labour rates	Low/base/high engineering allowance: RM 40 / 60 / 90 per hour. These are modelling rates, not salary survey findings or agency quotations.
Reuse assumption	A focused pilot reusing suitable existing software and devices. Actual code readiness has not been audited; hours must be rescoped after review.
Evidence codes	**P** = published provider price; **A** = explicit planning assumption; **Q** = provisional allowance requiring a quote. Published prices do not validate A or Q inputs.
Reserve and taxes	10% budget contingency on one-off and recurring subtotals. Not a statutory tax calculation. Confirm tax, FX/card fees and supplier terms before approval.

Budget versus accounting:*7 "CAPEX/setup" is a cash-planning bucket. Software and integration are capitalisation candidates only if the applicable recognition criteria are met. Launch, training, subscriptions and domain access are not automatically capital assets; an unspent reserve is not an incurred expense.

Hackathon versus live pilot: this is a deployment planning model, not the student team's actual expenditure. Existing devices and unpaid student effort can reduce cash outlay, but do not prove that professional delivery, maintenance or operator access are free.

One-off investment | delivery and setup

Engineering is calculated bottom-up from task hours. All hours and rates below are planning assumptions (A); operator-side onboarding is an unquoted allowance (Q).

Delivery item (A)	Hours L/B/H	Low RM	Base RM	High RM	Scope / basis
UX and existing-app review	8 / 12 / 20	320.00	720.00	1,800.00	Route, activity, wallet and redemption screens; reuse assessment.
Shade routing and map preparation	20 / 40 / 64	800.00	2,400.00	5,760.00	Public-data preparation, shade estimates and distance-aware route options.
GPS activity recording	16 / 24 / 40	640.00	1,440.00	3,600.00	Start/end recording, distance calculation and permission handling.
Points wallet and ledger	16 / 24 / 32	640.00	1,440.00	2,880.00	Credit once, balance/history, reservation and refund states.
Vending integration code	16 / 32 / 56	640.00	1,920.00	5,040.00	App connector, single-use redemption and vend-confirmation handling.
SDP planning dashboard	8 / 16 / 24	320.00	960.00	2,160.00	Consent-based route aggregation and shade-coverage display.
Privacy and access controls	8 / 16 / 24	320.00	960.00	2,160.00	Consent, role controls, retention and backup configuration.
QA and acceptance checks	16 / 24 / 40	640.00	1,440.00	3,600.00	Phone/GPS checks, duplicate credits, failures and ledger reconciliation.
Deployment and handover	8 / 12 / 16	320.00	720.00	1,440.00	Environment setup, release and concise operating documentation.
Software subtotal	116 / 200 / 316	**4,640.00**	**12,000.00**	**28,440.00**	Hours x RM 40 / 60 / 90. Base: 200 hours x RM 60.

What the base build buys: 200 task hours at RM 60/hour = RM 12,000. This is a planning allowance for a focused pilot, not a fixed-price offer, a timesheet or a statement that the current software has been audited. Existing components can lower hours; missing data, an incompatible machine or enterprise requirements can increase them.

One-off setup | consolidated funding requirement

Combine the software delivery subtotal with operator-side access, pilot preparation and the first year of domain access. Show the reserve separately.

One-off costs	Low RM	Base RM	High RM	Source / note
Software delivery subtotal (A)	4,640.00	12,000.00	28,440.00	From the itemised hours on page 3.
Operator onboarding / access setup (Q)	500.00	1,500.00	4,000.00	Operator-side configuration/permission fee; excludes our coding hours above.
Pilot launch and briefing (A)	150.00	300.00	600.00	Minimal signage, participant guidance and launch coordination.
Domain, first year paid upfront (A)	60.00	90.00	120.00	Annual allowance; no first-year domain cost is charged again in monthly OPEX.
Subtotal before reserve	5,350.00	13,890.00	33,160.00	Software + onboarding + launch + annual domain.
Setup contingency: 10% (A)	535.00	1,389.00	3,316.00	Reserve, not a supplier quotation or accounting asset.
TOTAL UPFRONT FUNDING	**5,885.00**	**15,279.00**	**36,476.00**	Planning cash requirement before monthly operation.

Base-case budget classification bridge

Bucket	Base amount	Interpretation
Development / integration candidates	RM 13,500	Software RM 12,000 + operator onboarding RM 1,500. Review recognition criteria and the actual nature of operator charges.
Other upfront cash costs	RM 390	Pilot launch RM 300 + annual domain RM 90. These are not automatically capital expenditure.
Unspent setup reserve	RM 1,389	10% of RM 13,890. A funding buffer, not an incurred asset or expense.
Total upfront funding	**RM 15,279**	**Cash-planning total used in the charts and combined summary.**

Existing-machine assumption: hardware purchase, machine rental, retrofit and new civil works are not included. This is conditional reuse, not a claim that every existing machine can accept our rewards. If the chosen site needs new equipment, obtain a separate quote before treating the pilot budget as complete.

OPEX | monthly operation and water rewards

Low/base/high estimate cost uncertainty at the SAME 100-participant / 200-bottle expected volume. They are not different adoption forecasts.

Cost line	Low RM	Base RM	High RM	Source / calculation basis	Status
App hosting: Vercel Pro*1	90.00	90.00	180.00	USD 20/month x RM 4.50; high includes a second deploying seat.	P + A
Database/auth: Supabase Pro*2	112.50	112.50	112.50	USD 25/month x RM 4.50; one Micro project covered by included compute credit.	P + A
Map tiles: MapTiler Flex*3	135.00	135.00	135.00	USD 30/month x RM 4.50; map services, not a separate shade-routing API.	P + A
Extra cloud/data usage	0.00	50.00	150.00	Allowance above included hosting/database/maps quotas; not a fixed vendor fee.	A
Notifications and monitoring	10.00	25.00	75.00	Allowance for email delivery, monitoring and operational messaging.	A
Software maintenance	160.00	480.00	1,440.00	4 / 8 / 16 hours per month x RM 40 / 60 / 90 per hour.	A
Programme administration	50.00	120.00	320.00	2 / 4 / 8 hours per month x RM 25 / 30 / 40: support and reconciliation.	A
Vending platform/access fee	0.00	100.00	300.00	Provisional recurring API/platform allowance. No fee schedule obtained.	Q
Drinking-water fulfilment*6	200.00	300.00	500.00	200 confirmed bottles/month x RM 1.00 / 1.50 / 2.50 all-in reward allowance.	A / Q
Subtotal before reserve	757.50	1,412.50	3,212.50	Sum of monthly recurring lines.	Calculated
Operating contingency: 10%	75.75	141.25	321.25	Allowance for unplanned operating variation.	A
EXPECTED MONTHLY OPEX	**833.25**	**1,553.75**	**3,533.75**	**Includes expected water rewards and operating reserve.**	Calculated
12-MONTH RUN-RATE	**9,999.00**	**18,645.00**	**42,405.00**	Assumes identical activity volume each month; not a growth forecast.	Calculated

Prevent double counting: Supabase's included compute credit covers one Micro project; no extra USD 10 is added. The first-year domain is paid in setup. Water unit cost assumes normal product and fulfilment margin; do not add a second refill margin. Remove the platform allowance if a final operator quote already bundles it.

Water economics | expected use and funded limits

Only drinking water is budgeted. No new machine purchase, snack rewards or other reward categories are included.

Budget driver	Formula / meaning	Base result
Expected monthly bottles	100 active participants x 4 maximum bottles x 50% redemption assumption	200 bottles
Base all-in unit allowance	Water + normal vending fulfilment; not an operator quote	RM 1.50/bottle
Expected monthly water spend	200 x RM 1.50	RM 300.00
Maximum monthly bottles at pilot cap	100 active participants x 4 bottles	400 bottles
Maximum monthly water spend	400 x RM 1.50	RM 600.00
Expected 3-month water spend	3 x RM 300	RM 900.00
Maximum 3-month water spend	3 x RM 600	RM 1,800.00

Volume sensitivity | base rates, same fixed operating capacity

Active participants (A)	Expected bottles / month	Water RM / month	OPEX RM / month	3-month funding RM
50	100	150.00	1,388.75	19,445.25
100	200	300.00	1,553.75	19,940.25
250	500	750.00	2,048.75	21,425.25

Unit-cost interpretation: base water fulfilment is RM 1.50/bottle. Total recurring programme cost per expected redemption is RM 7.77; all-in first-pilot cost per expected redemption is RM 33.23. These include software/support costs and are NOT the vending price of a bottle. The RM 1 ATLAS campaign*6 is context only, not our negotiated rate.

Combined summary | funding and spending schedule

A funding requirement is not profit. Supplier prices, usage, fees and final scope can change these planning totals.

Cost / period	Low RM	Base RM	High RM
Upfront setup funding	5,885.00	15,279.00	36,476.00
Expected monthly OPEX	833.25	1,553.75	3,533.75
Expected 3-month operating spend	2,499.75	4,661.25	10,601.25
Expected 3-month total	**8,384.75**	**19,940.25**	**47,077.25**
Expected annual operating run-rate	9,999.00	18,645.00	42,405.00
Year-one setup + 12 months	**15,884.00**	**33,924.00**	**78,881.00**

Base-case cash-planning schedule

Timing	Planned spend RM	Cumulative RM	What it covers
Before operation	15,279.00	15,279.00	Build, onboarding, launch, annual domain and setup reserve.
Operating month 1	1,553.75	16,832.75	Expected 200 water redemptions; includes operating reserve.
Operating month 2	1,553.75	18,386.50	Expected 200 water redemptions; includes operating reserve.
Operating month 3	1,553.75	19,940.25	Expected 200 water redemptions; includes operating reserve.

Funding for the full reward cap: at 400 monthly redemptions, base monthly OPEX becomes RM 1,883.75 and three-month funding becomes **RM 20,930.25**. This is RM 990.00 above the expected-use plan, including reserve. Treat this as the funded ceiling for the stated 100-person pilot cap, not an uncapped reward promise.

Expected value | the return we can defend

Keep the feasibility slide simple. Monetary ROI, payback and quantified ESG valuation require additional evidence and are not estimated here.

Expected value: repeat participation · traceable water rewards · evidence for pedestrian improvements

Expected value	Practical measure	Evidence boundary
Repeat participation	Share of first-time participants completing another activity within 7 days; completed activities per active user.	Measured during pilot; no improvement percentage assumed.
Traceable water rewards	Confirmed dispenses / redemption attempts; issued-code reconciliation; reward funds used.	200 expected bottles/month is a budget scenario, not an achieved outcome.
Pedestrian evidence	Frequently used route segments with low estimated shade; segments reviewed by SDP.	Candidate assessments; usage alone does not establish construction need.

Material items to resolve before committing funds

Decision	Required confirmation
Operator and site	Confirm water availability, route access hours, compatible hardware and API permission. No SDP machine has been verified.
Commercial terms	Replace RM 1,500 setup, RM 100/month platform and RM 1.50/bottle BASE allowances with operator terms; confirm fees, taxes and stock fulfilment.
Mapping and software	Audit existing app reuse and route-data coverage. MapTiler provides maps; detailed shade estimation still needs suitable geometry/data.
Scope boundary	New machine purchase/retrofit, paid survey or proprietary mapping, enterprise integrations and major legal/procurement requirements are not priced. Re-scope if required.

Business potential after shortlisting: a service or licensing model can be tested once a payer, contract scope and delivery cost are known. Sponsorship that reimburses water is funding, not automatically a profit margin. Repeat activity and planning evidence remain non-monetised benefits in this version.

Sources | assumptions and chart values

Asterisks connect provider-backed statements to their source. Sources checked 22 October 2026; public prices remain subject to change and terms.

Ref	Source	What it supports	Link
*1	Vercel Pro	USD 20/month with one deploying seat; extra seats and usage can add cost.	Open official source
*2	Supabase Pro	From USD 25/month; included compute credit covers one Micro project.	Open official source
*3	MapTiler Flex	USD 30/month plus overuse; commercial pilot map-service benchmark.	Open official source
*4	Silkron Vendron API	Third-party purchase validation and remote dispensing on Vendron machines; no public tariff used.	Open official source
*5	Silkron Code Redemption	Code/QR product redemption capability; not an SDP agreement.	Open official source
*6	ATLAS RM1 campaign	Mineral-water campaign benchmark only; not a current SDP/operator quote.	Open official source
*7	IFRS Foundation / IAS 38	Research versus qualifying development and intangible-asset recognition.	Open official source
*8	Mabe's SAC financial appendix	14 screenshots supplied by Adam on 22 October: format and assumptions discipline only.	User-supplied reference

Exact base-case chart values | copy into Canva

Chart	Bar values (RM)	Total RM
One-off setup	Software 12,000; operator onboarding 1,500; launch + domain 390; reserve 1,389.	15,279.00
Monthly OPEX	Cloud/maps/tools 412.50; maintenance/admin 600; platform/water 400; reserve 141.25.	1,553.75

Not externally validated: task hours, labour rates, FX rate, domain allowance, operator fees, water reimbursement, participant counts, redemption fraction and reserve are all explicit planning inputs. No operator quote, adoption study, profit forecast or measured ESG return is claimed. The appendix is reviewable cost modelling, not fabricated operating data.